Indian Meteorological Department Aviation Forecast Verification Web Portal with ML-Based Aerodrome Warning Prediction

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***Abstract*—Aviation weather forecasts are routinely checked against what was actually observed, but at the India Meteorological Department (IMD) this verification is largely manual, slow and difficult to scale. This paper presents a web portal that automates the verification of Take-Off Forecasts (TOF), Local Area Forecasts (LAF) and Aerodrome Warnings (ADWR) against METAR and upper-air observations using IMD tolerance criteria, and produces accuracy tables, graphs and downloadable reports. The portal is extended with a machine-learning module for aerodrome-warning support: two logistic regression classifiers estimate the probability of thunderstorm with rain (TSRA) in the next METAR and of a gust of 25 kt or more, and a linear regression model estimates wind speed, gust and direction. Models are trained on chronologically split METAR data (70/15/15) with F1-based threshold calibration. On 56,482 test samples the saved model reports 87.3% TSRA and 79.6% gust classification accuracy, with AUC of 0.89 and 0.84. The portal is hosted on IMD infrastructure and is in testing; its prediction output is decision support, not an official forecast.**

Keywords—aviation meteorology, forecast verification, METAR, aerodrome warning, TSRA, gust, logistic regression, linear regression, decision support

I. Introduction

A. Aviation Meteorology and Flight Safety

Aviation meteorology is the application of atmospheric science to the planning and safe conduct of air operations. Parameters such as surface wind, gusts, visibility, cloud, temperature, pressure (QNH) and significant weather such as thunderstorms directly influence take-off, climb, approach and landing. Meteorological service to air navigation is governed by ICAO Annex 3 [2], while the codes used to exchange observations and forecasts, including METAR, are defined in WMO Manual on Codes No. 306 [1]. Accurate and timely forecasts are therefore a critical input to flight planning and operational decision-making.

B. Indian Meteorological Department

The India Meteorological Department (IMD) is the national agency responsible for meteorological observations, forecasting and warnings in India [3]. Its aviation offices issue the Take-Off Forecast (TOF), Local Area Forecast (LAF) and Aerodrome Warning (ADWR). The quality of these products is assessed by comparing them against what was actually observed, a process known as forecast verification.

C. Problem with Manual Verification

Aviation forecast verification at IMD is largely manual, which makes it time-consuming, error-prone and difficult to scale. Forecast files and observation reports are available in different formats (text, PDF and CSV) and must be matched in time and compared parameter by parameter. The absence of an automated system delays performance evaluation and limits the feedback available for improving forecast accuracy.

D. Proposed Portal

The project was developed in coordination with senior IMD officials to remain aligned with operational requirements. It is a single web portal that automates verification of TOF, LAF and ADWR by comparing forecasts with METAR observations (for TOF and ADWR) and upper-air observations (for LAF), and generates accuracy tables, graphs and downloadable reports. The portal also includes a machine-learning Aerodrome Warning Prediction component: two logistic regression classifiers estimate the probability of TSRA (thunderstorm with rain) in the next METAR and of a gust of 25 kt or more, and a linear regression model estimates wind speed, gust and direction. The application is hosted on IMD infrastructure and is in the testing and validation phase.

E. Verification and Prediction

Forecast verification is retrospective: an issued forecast is compared with observations to measure its quality. Forecast prediction is prospective: historical METAR observations are used to estimate what is likely to occur next. Both rely on the same observation data, and together they let forecasters review past performance and consider a data-driven estimate of upcoming hazards.

F. Aim and Objectives

The aim is to develop a web portal that automates verification of IMD aviation forecasts against observations and supports meteorological personnel with machine-learning-based insights on aerodrome-warning-relevant weather. The objectives are:

- to automate verification of TOF, LAF and ADWR;
- to compare forecasts with METAR and upper-air observations using standard verification criteria;
- to generate verification reports, accuracy tables, graphs and performance metrics;

- to predict TSRA in the next METAR and gusts of 25 kt or above, and estimate wind speed, gust and direction by linear regression;
- to support forecasters in improving forecast reliability, with the prediction serving as decision support and not as an official forecast.

II. Literature Survey

A. Standards and Observations

ICAO Annex 3 specifies the meteorological service for international air navigation, including aerodrome forecasts, warnings and the observations they depend on [2]; IMD provides these services in India [3]. Murphy argues that forecast goodness should be judged on consistency, quality and value, motivating systematic verification [10]. METAR is the routine aerodrome weather report whose format is defined in [1]; it reports wind (with gusts), visibility, present weather (e.g., TS, RA, TSRA), cloud, temperature, dew point and QNH. METAR archives are available from OGIMET [4] and upper-air soundings from the University of Wyoming archive [5].

B. Verification Methodologies

Jolliffe and Stephenson describe the standard framework: matching forecasts to observations, forming contingency tables for categorical events and computing accuracy, hit rate and false-alarm measures, with error measures for continuous variables [8]. Murphy and Winkler's distributions-oriented view and the WMO/WWRP verification resources give further guidance [6], [7]. The portal applies tolerance-based criteria (e.g., ±30° wind direction) as used by IMD, a form of categorical hit/miss verification per parameter. Aerodrome warnings alert users to hazards such as thunderstorms and strong winds [2]; verifying them is an occurrence/non-occurrence problem within the validity period.

C. Machine Learning in Weather Prediction

Glahn and Lowry introduced Model Output Statistics, using regression to relate predictors to weather elements [11]. McGovern et al. review AI for real-time decisions on high-impact weather and stress that such tools should aid forecasters [12]. Wilks provides the statistical foundation for regression and probabilistic methods in atmospheric data [9]. Logistic regression models the probability of a binary outcome and suits TSRA and gust-exceedance prediction [13], [14]; linear regression models a continuous response and suits wind estimation [14]. Precision, recall, F1 and ROC analysis are standard for classifier evaluation [15].

D. Research Gap

Existing material largely treats verification and prediction separately. This work places both in one operational portal: automated verification of issued products and a supporting predictive module trained on historical METAR data.

III. Social Impact of the Problem

Weather is a recognised factor in aviation operations, and the reliability of aviation weather products affects the confidence of operational decisions; improved forecast quality ultimately benefits passengers, crew and airport users. Thunderstorms bring heavy rain, lightning, turbulence and sudden wind shifts, and strong gusts affect take-off and landing, particularly for light aircraft, so the reliability of aerodrome warnings is of operational importance. A warning is valuable only if issued in time and reliable; by scoring thunderstorm and gust/wind warnings separately, the portal highlights where warnings need to improve. Automation also removes repetitive manual comparison, reduces transcription errors and applies the same criteria every time. The ML module offers probabilistic estimates that forecasters may consider alongside their own analysis; it is a decision-support aid only, does not control flights, does not issue official warnings and does not guarantee flight safety.

IV. System Design

A. Overall Architecture

The portal follows a pipeline architecture (Fig. 1). A user selects an ICAO station and a UTC period; the portal accepts the forecast files (TOF text, LAF PDF, ADWR text) and observation data (METAR reports, upper-air CSV); the Verification Engine performs time matching, parameter comparison and accuracy calculation; and results are presented as accuracy percentages, tables, graphs and downloadable reports. The ML prediction module is added alongside the engine and uses historical METAR observations.

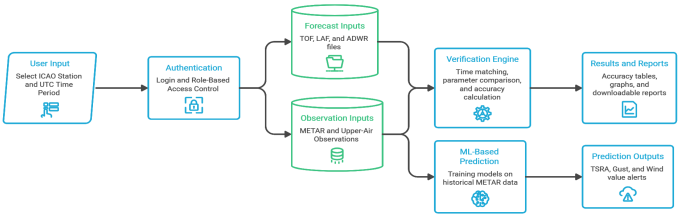


Fig. 1. System architecture of the web portal.

B. Role-Based Access Control

Users log in, after which role-based access control (RBAC) directs them to one of three roles (Fig. 2): Super Admin (full system management), Admin (user management and user logs) and User (aviation verification dashboard).

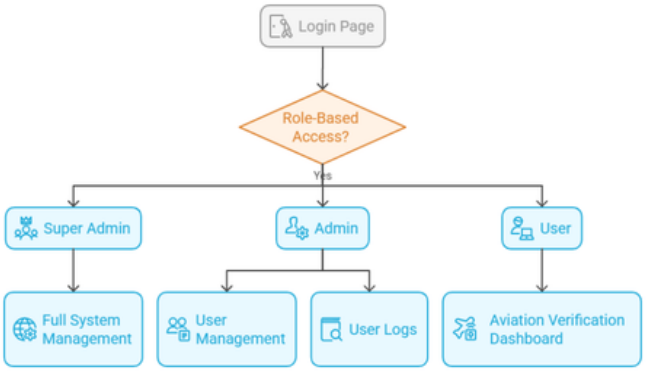


Fig. 2. Role-based access control (RBAC) workflow.

C. Functional Modules

The portal has four modules (Table I). The TOF module compares TOF text files with METAR observations; the LAF module compares LAF PDF reports with upper-air observations; the ADWR module compares warning files with METAR for thunderstorm occurrence and wind/gust exceedance; and

the prediction module uses historical METAR data to estimate TSRA and gust probabilities and wind values. It contains two logistic regression classifiers (TSRA in the next METAR: gust ≥ 25 kt) and one linear regression model (wind speed, gust, direction).

TABLE I. SUMMARY OF PORTAL MODULES

Module	Forecast / Observation input	Outputs
TOF	TAKE-OFF text / METAR	Accuracy tables, daily graphs
LAF	LAF PDF / upper-air CSV	Parameter-wise accuracy, IMD-format report
ADWR	Warning text / METAR	Warning-wise accuracy, tables, summary cards, graphs
Prediction	Historical METAR / next METAR	TSRA and gust probability, wind estimates

D. Data Flow and Training Workflow

Forecast files and observations enter the verification path, which outputs accuracy results; historical METAR observations enter the ML path, which outputs probabilities and wind estimates. METAR is common to both: it is the ground truth for TOF and ADWR verification and the training data for the ML module. The dataset is divided chronologically into 70% training, 15% calibration and 15% testing data; models are fitted on the training set, the alert threshold is chosen on the calibration set by maximising F1, and final performance is reported on the untouched test set. Chronological splitting is used because random shuffling would leak future information into training.

E. Integration of Verification and Prediction

Verification tells forecasters how reliable past products were; prediction estimates what may happen in the next observation. Verification can reveal weak areas (e.g., the lower gust/wind warning accuracy in the sample result), and the ML module targets exactly these hazards. Both share the METAR data foundation.

V. FLOW CHARTS

Fig. 3 shows the overall portal workflow: login with credential check, role check (Super Admin to system management, Admin to user management and logs, User to module selection among TOF, LAF, ADWR and ML prediction), selection of ICAO station and UTC period, upload or loading of forecast and observation data, processing and display of results. Fig. 4 shows the verification workflow: input, parsing and time matching, observation-found check, tolerance check, recording of a hit or miss, accuracy computation and report generation.

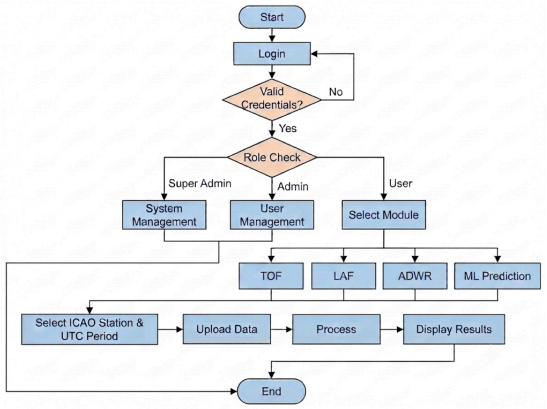


Fig. 3. Overall portal workflow.

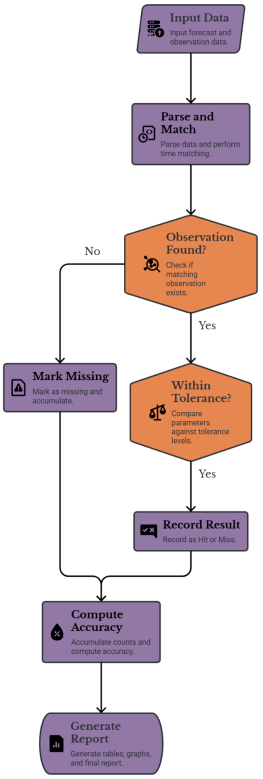


Fig. 4. Forecast verification workflow.

ML workflow: historical METAR data are split chronologically 70/15/15, logistic regression (TSRA, gust ≥ 25 kt) and linear regression (wind) models are trained, probabilities on the calibration data are used to select the F1-maximising threshold, the test data are evaluated, and an alert is raised when the probability meets the threshold.

VI. SYSTEM IMPLEMENTATION

A. Web Application

The portal is a web application hosted on IMD infrastructure with a login page, role-based dashboards and a verification dashboard. The frontend uses HTML, CSS and

JavaScript; the backend uses Flask (Python); the models use scikit-learn; and structured data is stored in SQLite (Table II). The deployment status is testing and validation.

TABLE II. IMPLEMENTATION TECHNOLOGY STACK

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Flask (Python)
Database	SQLite
ML library	scikit-learn
ML models	Logistic, Linear Regression
Hosting	IMD infrastructure

B. Data Processing and Inputs

The user selects an ICAO station and UTC period. Inputs are TAKE-OFF text files and METAR reports (TOF; Fig. 5), LAF PDFs and upper-air observations in CSV form (LAF; Fig. 6), and Aerodrome Warning text files with METAR reports (ADWR; Fig. 7). Fig. 7 also shows the prediction form, in which the station, date, time (UTC) and observation lookback are entered.

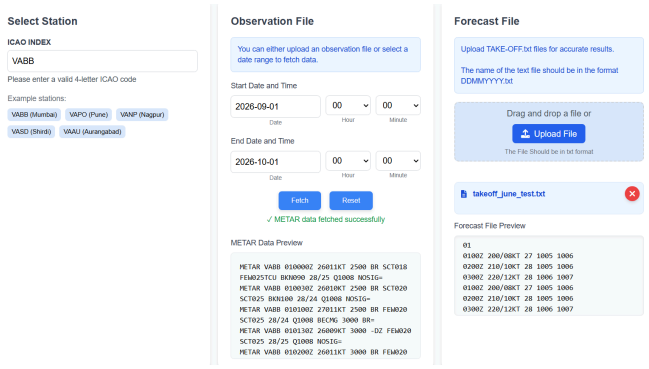


Fig. 5. Sample TOF text and METAR observation inputs.

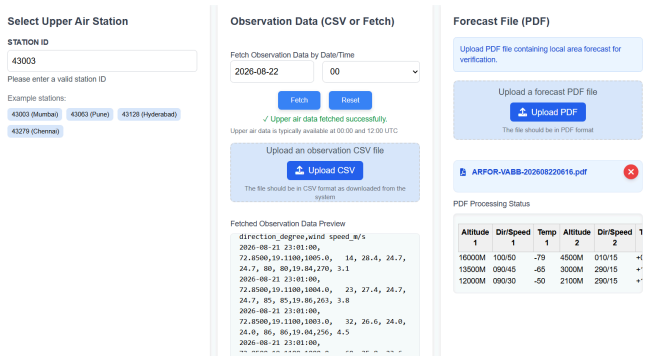


Fig. 6. Sample LAF report and upper-air observation inputs.

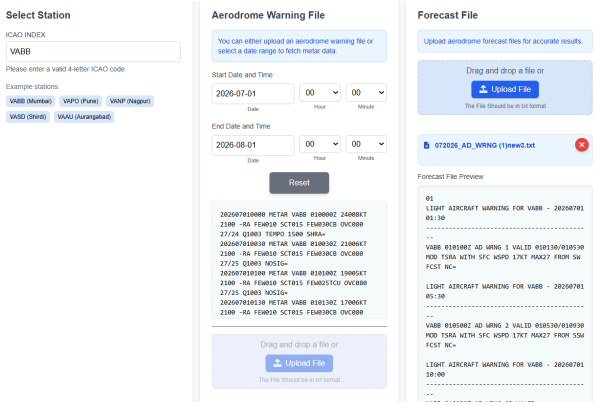




Fig. 7. Sample Aerodrome Warning and METAR inputs, with the prediction form.

C. METAR Parsing and Comparison

Each METAR line contains observation time, station, wind direction and speed (with optional gust group such as G20KT), visibility, present weather, cloud groups, temperature/dew point and QNH (e.g., Q1005). The Verification Engine extracts these fields, matches each forecast time to the corresponding observation and compares each parameter against its tolerance. Upper-air wind speed is given in m/s and is converted to knots (1 m/s ≈ 1.944 kt) before comparison.

D. Verification Criteria

Tolerances for TOF and LAF are given in Tables III and IV. Thunderstorm warnings are verified by checking for TS or TSRA in the METAR observations during the warning validity; gust/wind warnings are verified against the 25 kt threshold used across the portal.

TABLE III. TOF VERIFICATION CRITERIA

Parameter	Tolerance
Wind direction	±30°
Wind speed	±5 knots
Temperature	±1 °C
QNH	±1 hPa

TABLE IV. LAF VERIFICATION CRITERIA

Parameter	Criterion
Temperature	±2 °C
Wind speed	±10 knots
Wind direction	±30°
Weather	Occurrence / non-occurrence

VII. ML PREDICTION MODELS

A. Preprocessing and Chronological Splitting

Historical METAR observations are ordered by time and split into the first 70% (training), next 15% (calibration) and final 15% (testing). The test set contains 56,482 samples. Based on the 15% test split, this implies a total dataset of approximately 376,546 observations.

B. Logistic Regression

For a feature vector x, the event probability is

$$P(y=1|x) = 1 / (1 + e^{-(w \cdot x + b)})$$

where w and b are learned weights. Two separate classifiers are used: TSRA in the next METAR observation, and gust ≥ 25 kt. A

probability above the chosen threshold raises an alert.

C. Linear Regression

A separate linear model, $\hat{y} = w \cdot x + b$, estimates wind speed, wind gust and wind direction.

D. Calibration-Based Threshold Selection

Rather than the default 0.5 cut-off, the alert threshold is chosen on the calibration set to maximise F1 = 2·Precision·Recall/(Precision + Recall), balancing missed events against false alarms, which matters when events such as TSRA are rare.

E. Prediction Output

The module outputs event probabilities, the alert decision for the selected threshold and estimated wind values, presented in the portal as decision support beside the verification dashboard.

VIII. Results and Analysis

A. Forecast Verification Results

The portal produces parameter-wise and overall accuracy, tables, graphs and reports, reducing manual effort. Fig. 8 shows the TOF verification report for VABB from 01/09/2025 to 30/09/2025 with daily wind direction, wind speed, temperature, QNH and overall accuracy. Fig. 9 shows the LAF result for station 43003 (VABB) on 20/04/2025, 14:00–22:00 UTC: 50% temperature, 100% wind speed, 85.71% wind direction and 0% weather accuracy. Fig. 10 shows ADWR verification for VABB for September 2025: thunderstorm warning accuracy 91% and gust/wind warning accuracy 35%.

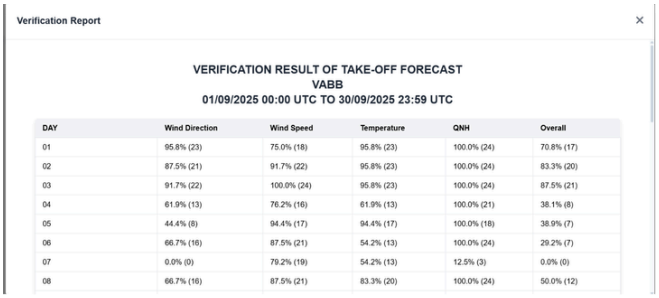


Fig. 8. Results of Take-Off forecast verification, VABB, 01/09/2025 to 30/09/2025.

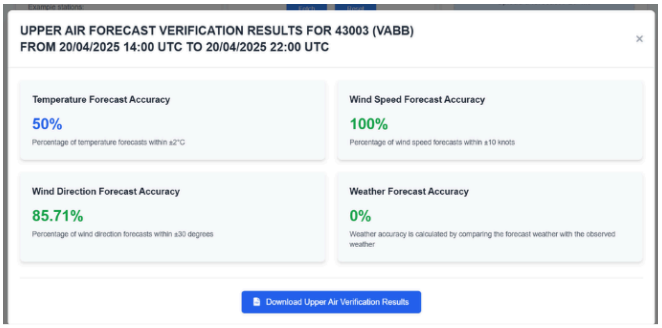


Fig. 9. Results of Local Area forecast verification, station 43003 (VABB), 20/04/2025 14:00–22:00 UTC.

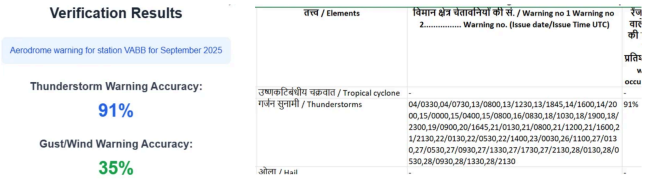


Fig. 10. Results of Aerodrome Warning verification.

B. Aerodrome Warning Prediction Results

Table V gives results for the currently saved model. Fig. 11 shows a sample predicted warning, labelled as experimental and not an official IMD warning. Table VI gives the linear regression metrics on the test set.

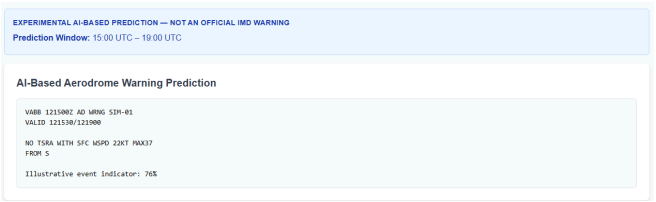


Fig. 11. Predicted Aerodrome Warning result.

TABLE V. Reported Prediction Results (Saved Model)

Metric	Value
TSRA classification accuracy	87.3%
Gust (≥25 kt) accuracy	79.6%
Test samples	56,482
Precision	77%
Recall	80%
F1 score	78.5%

C. Linear Regression Evaluation

TABLE VI. Linear Regression Metrics on the Test Set

Parameter	R^2	MAE	RMSE
Wind speed	0.68	0.18	0.24
Wind gust	0.46	0.27	0.35
Wind direction	0.34	0.21	0.29

D. Significance and Limitations

The classifiers perform well above chance on a chronologically separated test set, indicating useful predictive signal. The TSRA classifier achieved an AUC of 0.89 and the gust classifier 0.84; the F1-optimal thresholds on the calibration set were 0.41 (TSRA) and 0.47 (gust). Limitations remain: station and period coverage are limited, the system is in testing at IMD, and the ML output is a statistical estimate to be interpreted by forecasters, not an official forecast.

IX. Conclusion

This work delivered a web-based Aviation Forecast Verification Portal that automates verification of Take-Off Forecasts, Local Area Forecasts and Aerodrome Warnings against METAR and upper-air observations using the stated criteria. It produces parameter-wise and overall accuracy, tables, graphs and reports, reducing manual effort and ensuring consistent evaluation.

The portal was extended with an ML-based Aerodrome Warning Prediction module: two logistic regression classifiers for TSRA and gusts of 25 kt or more, and a linear regression model for wind speed, gust and direction. With a

chronological 70/15/15 split and F1-based threshold calibration, the saved model reports 87.3% TSRA and 79.6% gust accuracy on 56,482 test samples, with AUC of 0.89 and 0.84.

Future work includes evaluation across more stations and seasons, richer predictors, and automated verification of the ML predictions through the portal. The portal is a decision-support tool and must not be treated as an autonomous replacement for official forecasting and warning procedures.

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